Arman Motiwala

+1(617)-223-7071 | Boston, MA | armanmotiwala02@gmail.com | linkedin.com/in/arman-motiwala02/ | armanmotiwala.vercel.app

Summary

Graduate AI researcher (MS, Northeastern) specializing in multimodal ML, LLM fine-tuning, and computer vision. IEEE-published author with production experience in deep learning systems, model evaluation pipelines, and privacy-aware AI. Passionate about advancing state-of-the-art machine learning algorithms to deliver innovative, real-world user experiences.

Education

Northeastern University, Khoury College of Computer Sciences — Boston, MA
Master of Science in Artificial Intelligence — *Sep 2025 – Dec 2027*

Mukesh Patel School of Technology Management and Engineering — Mumbai, India
Bachelor of Technology in Artificial Intelligence — *Jul 2021 – May 2025*

Technical Skills

Languages: Python, Swift (familiar), SQL, JavaScript, Java
ML Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, CoreML
Deep Learning: CNNs, Transformers, Diffusion Models, U-Net++, EfficientNet, LoRA, LLM Fine-Tuning, Reinforcement Learning
Generative & Multimodal AI: RAG, Prompt Engineering, Vision-Language Models, Multimodal LLMs, Vector Search
Research: Ablation Studies, Benchmark Design, Model Evaluation, Statistical Analysis, Academic Publishing
Deployment: FastAPI, Docker, Vertex AI, Google Cloud Platform, OpenAI API

Experience

KNOW LOCTOC PTE. LTD. — Singapore (Remote)
Software Developer Intern — *Jan 2025 – Jul 2025*

- Engineered a multimodal AI evaluation system (Vertex AI, GPT-4o) to benchmark visual-text alignment across 1,000+ product images, achieving 95% accuracy through iterative prompt optimization and model calibration.
- Architected an NLP feedback pipeline (FastAPI, Milvus Vector DB) with multi-class sentiment classification across 5 categories (F1: 0.87, ROC-AUC: 0.91), cutting manual audit workload by 90%.
- Developed a RAG-based adaptive quiz system integrating OpenAI LLMs with vector search, generating 6 Bloom's Taxonomy-aligned question types with adversarial prompt robustness testing.

Sequel String AI — India (Remote)
AI/ML Intern — *Apr 2024 – Jul 2024*

- Automated a computer vision pipeline for utility meter digit extraction (OpenCV, REST APIs), processing 500+ daily images and improving structured data throughput by 40%.
- Incorporated neural machine translation and language detection modules, expanding multilingual accessibility across 3 regional languages and enhancing downstream model usability.

Projects

Offline Reinforcement Learning: CQL vs. Baselines on D4RL | *RL Research* — Jan 2025 – Present

- Implemented Conservative Q-Learning (CQL) alongside BCQ and TD3+BC on D4RL benchmarks (HalfCheetah, Hopper) using PyTorch, examining distributional shift and policy generalization.
- Executed systematic ablation studies on conservatism hyperparameters and reward shaping, quantifying performance gaps across offline RL baselines.

Medical Image Analysis: Fracture Detection & Chest X-Ray Classification | *Deep Learning* — Oct 2025 – Dec 2025

- Designed a U-Net++ segmentation framework (EfficientNet-B4 encoder, SCSE attention) for fracture localization, attaining 61.48% Dice score and surpassing YOLOv8s-seg baseline by 4.4%.
- Trained a 4-class chest X-ray classifier via EfficientNet-B3 transfer learning (84.92% accuracy, 0.84 macro-F1, 3× fewer parameters than VGG-16); formulated a custom Dice-Focal-BCE hybrid loss to resolve extreme class imbalance (<2% positive pixels).

Auto-Assess: NL-to-SQL Platform | *IEEE SCEECS 2025* — Jul 2024 – Nov 2024

- Fine-tuned Llama 3.1 via LoRA on the Spider dataset for natural language to SQL translation, reaching 87% execution accuracy on WikiSQL; research published at IEEE SCEECS 2025.
- Established an end-to-end LLM evaluation pipeline assessing execution accuracy and logical form correctness, leveraging Hugging Face Transformers, PyTorch, MongoDB, and Flask APIs.

Research & Certifications

Publication: M. A. Motiwala, A. Joshi, and R. Chapaneri, "Auto-Assess: Automated Evaluation Platform for Enhanced SQL Learning," *IEEE SCEECS 2025*, Bhopal, India. [IEEE Xplore]
Certifications: AI Agents Using RAG and LangChain (IBM), Multimodal RAG via Vertex AI (Google), Machine Learning with Python (IBM)
Achievements: Finalist – Datathon 2.0 (2024), Finalist – Cyber Cypher 3.0 Advanced Level (2024)
Research Interests: Large Language Models, Multimodal AI, Diffusion Models, Reinforcement Learning, Privacy-Preserving ML